

SIMATS ENGINEERING
CO3-ASSESSEMNT TOOL-3
CSA0302-DATA STRUCTURES

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9.

44

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22 66

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11 33 55 77

A. Binary Tree (4 marks): Write the preorder, inorder, postorder and level-order traversals of the given tree.

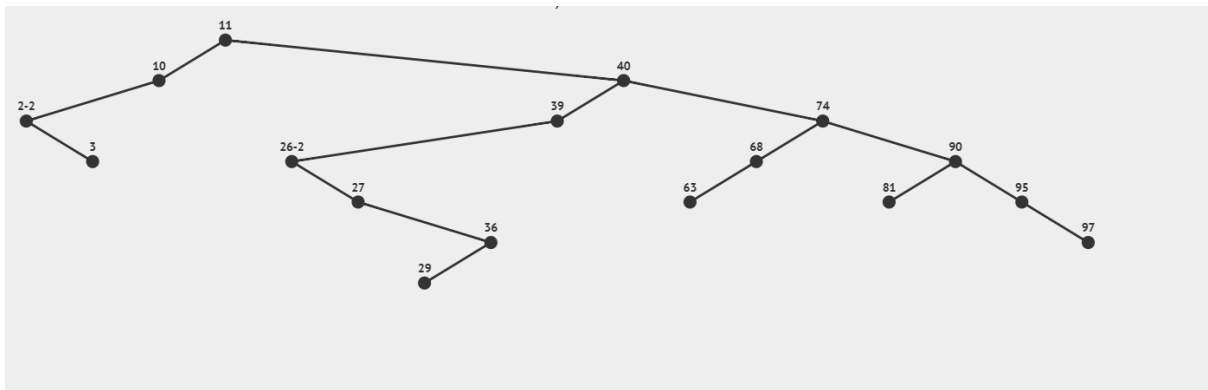
B. BST (3 marks): Insert the keys in this order: 44, 22, 66, 13, 31, 57, 79. Draw the final BST and write its postorder traversal.

C. AVL Tree (3 marks): Insert 90, 70, 50 into an empty AVL tree. Name the rotation, draw the balanced tree and write its preorder traversal.

A.

- **Preorder:** $A \rightarrow B \rightarrow D \rightarrow E \rightarrow C$
- **Inorder:** $D \rightarrow B \rightarrow E \rightarrow A \rightarrow C$
- **Postorder:** $D \rightarrow E \rightarrow B \rightarrow C \rightarrow A$
- **Level-order:** $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$

B.



C.

